1) Project Title

Design and Development of a Full-Stack AI-Based Public Infrastructure Accident Reporting and Compensation Management System

2) Statement About the Problem

Public infrastructure such as roads, bridges, footpaths, drainage systems, streetlights, and construction zones often suffer from poor maintenance. Accidents caused by potholes, open drains, broken pavements, or unsafe public structures are common. However, victims face difficulty in **reporting incidents**, **providing proof**, and **tracking compensation claims** due to slow, manual, and unorganized government processes.

Currently, there is no unified digital platform that allows citizens to report infrastructure-related accidents, validate claims, and ensure transparency in compensation handling. This results in delayed justice, lack of accountability, and repeated accidents at the same locations.

3) Why Is This Particular Topic Chosen

- Increasing number of **road and infrastructure-related accidents**
- Strong relevance to **smart city and e-governance initiatives**
- Direct **social impact and public safety improvement**
- Lack of transparent accident reporting systems
- Fits well with **AI + Full-Stack web development**
- Easy to justify and demonstrate at **BCA level**

4) Objective and Scope

Objectives

- To provide a digital platform for reporting public infrastructure accidents
- To analyze accident data and identify high-risk locations
- To assist authorities in managing compensation claims efficiently
- To improve transparency and accountability in public infrastructure maintenance
- To reduce repeated accidents through data-driven insights

Scope

- Covers accidents caused by public infrastructure defects
- Web-based reporting and tracking system
- AI-based validation and pattern analysis
- Dashboard for authorities and administrators
- Prototype-level academic implementation

5) Methodology

1. **Requirement Analysis**
   - Study common public infrastructure accident scenarios
   - Identify stakeholders (citizens, authorities)
2. **System Design**
   - Design frontend, backend, database, and AI workflow
3. **Accident Reporting Module**
   - Capture accident details, location, and description
   - Upload evidence such as images or documents
4. **AI-Based Analysis**
   - Classify accident types
   - Detect accident-prone locations using data patterns
5. **Compensation Tracking**
   - Track claim status and processing stages
6. **Testing & Validation**
   - Test accuracy, usability, and system reliability

6) Hardware & Software Requirements

Hardware Requirements

- Laptop / Desktop Computer
- Internet connection
- Minimum 4GB RAM

Software Requirements

- Frontend: HTML, CSS, JavaScript
- Backend: Python (Django / Flask)
- Database: MySQL / SQLite
- AI/ML: Python, Scikit-learn (basic classification models)
- Tools: VS Code
- OS: Windows / Linux

7) Contributions of the Project to Society

- Enables **quick and transparent accident reporting**
- Improves public safety awareness
- Supports government authorities in decision-making
- Helps reduce repeated accidents at dangerous locations
- Encourages citizen participation in governance
- Builds trust through transparent compensation tracking

8) Schedule of the Project

Phase	Activity	Duration
Phase 1	Problem study & requirement analysis	1 Week
Phase 2	System design & database modeling	1 Week
Phase 3	Accident reporting & claim modules	2 Weeks

Phase 4	AI analysis & dashboard development	3 Weeks
Phase 5	Integration & testing	2 Weeks
Phase 6	Documentation & final submission	1 Week

Project Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10
Problem Study & Requirement Analysis	✓	✓								
System Architecture & Database Design		✓	✓							
Citizen Accident Reporting Module			✓	✓						
Evidence Upload & Claim Registration				✓	✓					
AI-Based Accident Classification & Risk Analysis					✓	✓	✓			
Authority Dashboard & Compensation Tracking						✓	✓			
System Integration & Functional Testing							✓	✓		
Performance Evaluation & Improvements								✓	✓	
Documentation & Final Submission									✓	✓

9) Limitations of the Project

- AI predictions depend on data availability
- Real-time verification may not be possible
- Legal compensation rules vary by region
- Prototype does not cover large-scale deployment
- Requires internet access for reporting

10) References

1. Ministry of Road Transport and Highways – Government of India
2. Smart Cities Mission – Official Guidelines
3. Django Official Documentation
4. Machine Learning with Python – Scikit-learn
5. IEEE Research Papers on Smart Governance Systems